

Rectangular matrix multiplication from shared-leg entropy

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Abstract

In this note, we extend the matrix-multiplication analysis in [8] to rectangular products and prove

$$\omega(1, k, 1) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases}$$

In particular, $\omega(1, \frac{1}{2}, 1) = 2$ and the dual exponent satisfies $\alpha \geq \frac{1}{2}$. A Hermite-interpolation degeneration gives ordinary homogeneous limits of convolution character profiles. The shared-leg entropy inequality and degenerations of convolution tensors from [8] constrain their slopes and intercepts, yielding $\sum_i \arccos \sqrt{p_i} \geq \pi/2$ for the three dot-product exponents of every tensor character. Tensor-spectrum duality then gives both the rectangular curve and a bound for $\omega(a, b, c)$, with exact exponent $a + b$ whenever $a \geq b \geq c \geq 0$ and $c(a + b) \leq ab$. As an application, Zwick’s algorithm [12] for all-pairs shortest paths in directed unweighted graphs runs in $O(n^{2.5})$ time. Combining the rectangular bound with the $(\min, +)$ -product improvement of Alman and Vassilevska Williams [1] further gives $O(n^{2.4999})$ running time.

1 Introduction

The motivation for this note is to gather, in one place, useful bounds and algorithmic consequences of the groundbreaking matrix-multiplication result in [8], with a focus on rectangular products.

Complexity convention. Throughout this note, all complexity bounds suppress a $+o(1)$ term in the exponent: $O(n^c)$ denotes $n^{c+o(1)}$. All shape parameters are fixed.

Rectangular matrix multiplication is a basic ingredient in algorithms for all-pairs problems, including shortest paths, lowest common ancestors, and Hamming distance. Its exponents depend on the shape of the product; an improvement for square matrices need not give a comparable improvement for every rectangular shape. The analysis in [8] establishes the square bound $\omega \leq 2.25$ using a shared-leg entropy inequality and degenerations of convolution tensors. We extend that analysis to rectangular products by retaining the orientation information lost under full leg symmetrization.

We work over $\mathbb{C}$. By invariance of asymptotic rank under field extension, the resulting matrix-multiplication exponent bounds also hold over every field of characteristic zero [2, Remark 1.3].

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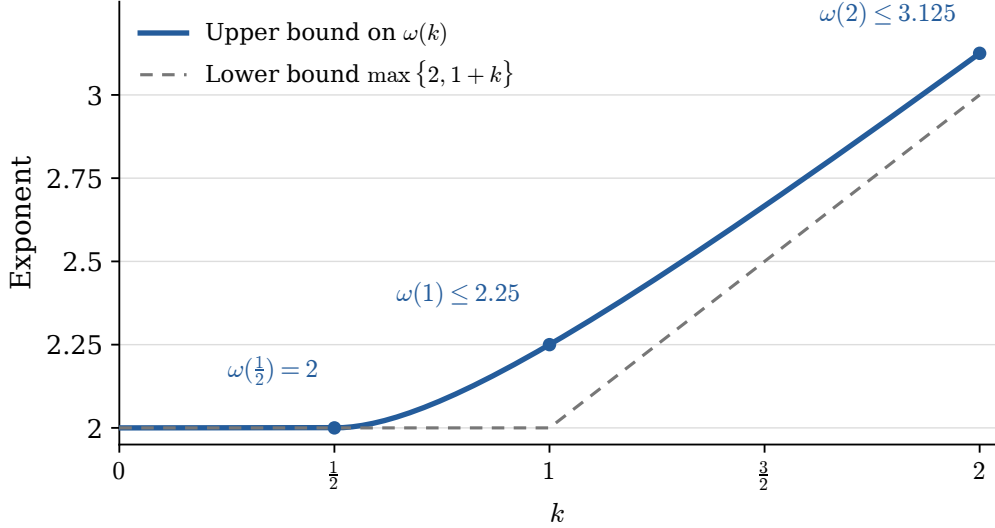


Figure 1: The upper bound on $\omega(k)$ from Theorem 1.1 for $0 \leq k \leq 2$, together with the size lower bound $\max\{2, 1+k\}$. The bounds coincide for $0 \leq k \leq \frac{1}{2}$.

For fixed nonnegative real u, v, w , let $\omega(u, v, w)$ be the infimum of the exponents e for which multiplying an $\lceil n^u \rceil \times \lceil n^v \rceil$ matrix by a $\lceil n^v \rceil \times \lceil n^w \rceil$ matrix uses $O(n^e)$ arithmetic operations, with the convention above. Write $\omega(k) = \omega(1, k, 1)$ and $\alpha = \sup\{k \geq 0 : \omega(k) = 2\}$.

Theorem 1.1. *For every fixed real $k \geq 0$,*

$$\omega(k) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases} \quad (1)$$

Consequently $\omega(k) = 2$ for $0 \leq k \leq \frac{1}{2}$, $\alpha \geq \frac{1}{2}$, and $\omega(2) \leq 3.125$.

Theorem 1.1 gives exponent two for inner dimension up to $n^{0.5}$, and an upper bound within $\frac{1}{4k}$ of the input-size lower bound $1+k$ for $k \geq 1$.

Our proof bounds the dot-product exponents p_i of every tensor character, defined in (2). Hermite interpolation splits a convolution tensor into a direct sum of smaller convolutions. This makes each character profile superadditive, so its ordinary scaling limit exists and is homogeneous of degree one. The shared-leg inequalities force the powered limits to have common slopes and bounded intercepts. Coupling the intercepts of all three orientations gives $\sum_i \arccos \sqrt{p_i} \geq \pi/2$. Optimizing this constraint gives the general bound in Corollary 5.1; Theorem 1.1 follows by setting two dimension exponents equal.

Relation to the preceding analysis. The shared-leg entropy inequality, determinant construction, and three-sector degeneration in Lemma 2.1 are inherited from Corollary 3.2 and the proofs of Lemmas 4.1 and 4.2 of [8]. Their oriented powered-concavity and tripling inequalities in Lemma 2.2 follow directly from those inputs.

The additional steps here are the Hermite direct-sum degeneration (Lemma 3.1), its use with the standard superadditive-limit argument (Lemma 3.2), and the common-slope and intercept analysis (Lemma 4.1). Coupling the six intercept ratios gives the angular constraint in Proposition 4.2. These steps extend the preceding analysis while using standard interpolation, concavity, and optimization tools; we make no priority claim for those tools or for the Hermite construction itself. Section 5 optimizes the constraint and applies standard

tensor-spectrum duality. Section 6 substitutes the resulting bounds into known algorithmic tradeoffs.

2 Tensor characters and convolution

We view a tensor as a trilinear form in three groups of variables, called its *legs*. A *restriction* applies linear substitutions independently on the three legs; a *degeneration* is a limit of such restrictions. Here the substitutions may depend on $\varepsilon \neq 0$; each coefficient of the resulting tensor converges as $\varepsilon \rightarrow 0$, even if the substitutions themselves do not.

A *tensor character* assigns each tensor T a nonnegative real number $\lambda(T)$, with

$$\lambda(xyz) = 1, \quad \lambda(S \oplus T) = \lambda(S) + \lambda(T), \quad \lambda(S \otimes T) = \lambda(S)\lambda(T).$$

Here $\oplus$ uses disjoint variable groups and $\otimes$ is the Kronecker product. A character is also monotone: if T restricts to or degenerates to S , then $\lambda(S) \leq \lambda(T)$. The collection of these characters is the *tensor spectrum*. For the oriented dot-product tensors

$$B_X(q) = x \sum_{j=1}^q y_j z_j,$$

and its leg permutations $B_Y(q), B_Z(q)$, define the *dot-product exponents* p_i of λ , for $i \in \{X, Y, Z\}$, by

$$\lambda(B_i(q)) = q^{p_i}, \quad 0 \leq p_i \leq 1. \quad (2)$$

The *rank* of a tensor is the minimum number of products of three linear forms, one in each leg, whose sum equals that tensor. Characters are positive on nonzero tensors and bounded above by rank. The matrix-multiplication tensor $\langle A, B, C \rangle = \sum_{u,v,w} x_{uv} y_{vw} z_{wu}$ has character value

$$\lambda(\langle A, B, C \rangle) = A^{p_Y} B^{p_Z} C^{p_X}, \quad (3)$$

by its factorization into three oriented dot products. Define the asymptotic rank by $\widetilde{\text{rank}}(T) = \lim_{m \rightarrow \infty} \text{rank}(T^{\otimes m})^{1/m}$. We use the standard duality

$$\widetilde{\text{rank}}(T) = \max_{\lambda} \lambda(T), \quad (4)$$

as stated, for example, in [3, Section 4.1, p. 18]. Thus it suffices to upper-bound the value of every character on the relevant matrix-multiplication tensor. Section 5 converts the resulting asymptotic-rank bound into a running-time exponent by taking tensor powers.

For distinct tensor legs i, j , define the *convolution tensor*

$$C_{ij}(r, s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_{\ell}^{(j)} x_{h+\ell}^{(k)}$$

on input legs i, j and the remaining output leg k . It represents multiplication of two polynomials through convolution of their coefficient vectors. Its rank is $r + s - 1$ over $\mathbb{C}$: interpolation at $r + s - 1$ distinct points gives the upper bound, and the output flattening gives the lower bound. A smaller pair of input lengths gives a restriction.

We call the blocks T_h *matched sectors sharing the X-leg* if $T = \sum_h T_h$ and the blocks may share X -variables, but use pairwise disjoint sets of Y -variables and pairwise disjoint sets of Z -variables.

The following results restate Corollary 3.2 and the constructions in the proofs of Lemmas 4.1 and 4.2 of [8, pp. 5–8].

Lemma 2.1 ([8]). *The following statements hold for every leg permutation.*

1. *If a tensor has matched sectors T_h sharing leg i , with the sectors disjoint on the other two legs, then for every probability vector q ,*

$$\lambda(T) \geq \exp(p_i H(q)) \prod_h \lambda(T_h)^{q_h}, \quad H(q) = - \sum_h q_h \log q_h. \quad (5)$$

The same inequality applies to a degeneration onto such sectors.

2. *The determinant construction for $C_{ij}(r, s) \otimes B_i(2)$, recapped in Appendix A, yields for $s \geq 2$ two matched sectors $C_{ij}(r, s+1)$ and $C_{ij}(r, s-1)$ sharing leg i .*
3. *For $r, s \geq 1$, $C_{ij}(r, 3s+r-1)$ degenerates to three matched sectors sharing leg i : two copies of $C_{ij}(r, s)$ and one copy of $C_{ik}(r, s)$, where k is the remaining leg.*

2.1 Optimizing the shared-leg inequality

Fix a character λ . For $p_i > 0$ and nonzero matched sectors T_h sharing leg i , put $a_h = \lambda(T_h)^{1/p_i}$ and $A = \sum_h a_h$. For every positive probability vector q , weighted AM–GM gives

$$e^{H(q)} \prod_h a_h^{q_h} = \prod_h \left(\frac{a_h}{q_h} \right)^{q_h} \leq \sum_h q_h \frac{a_h}{q_h} = A.$$

Equality holds at $q_h^* = a_h/A$. Substituting this maximizing vector into (5) yields

$$\begin{aligned} \lambda(T)^{1/p_i} &\geq e^{H(q^*)} \prod_h a_h^{q_h^*} = \prod_h A^{q_h^*} \\ &= A = \sum_h \lambda(T_h)^{1/p_i}. \end{aligned} \quad (6)$$

2.2 Oriented convolution profiles

A *profile* here is a function of the two input lengths. For distinct legs i, j , define

$$f_{ij}(r, s) = \lambda(C_{ij}(r, s)) \quad (r, s \geq 1). \quad (7)$$

These profiles are nondecreasing in each coordinate, and the rank bound gives

$$1 \leq f_{ij}(r, s) \leq r + s - 1. \quad (8)$$

Interchanging the two inputs gives the same tensor, so $f_{ij}(r, s) = f_{ji}(s, r)$.

Lemma 2.2 (Powered concavity and tripling). *For $p_i > 0$, the sequence $s \mapsto f_{ij}(r, s)^{1/p_i}$ is concave, and for all integers $r, s \geq 1$,*

$$f_{ij}(r, 3s+r-1)^{1/p_i} \geq 2f_{ij}(r, s)^{1/p_i} + f_{ik}(r, s)^{1/p_i}, \quad (9)$$

where i, j, k are distinct. For $p_j > 0$, the sequence $r \mapsto f_{ij}(r, s)^{1/p_j}$ is also concave.

Proof. The determinant construction and (6) give, for $s \geq 2$,

$$\begin{aligned} 2f_{ij}(r, s)^{1/p_i} &= \lambda(C_{ij}(r, s) \otimes B_i(2))^{1/p_i} \\ &\geq f_{ij}(r, s+1)^{1/p_i} + f_{ij}(r, s-1)^{1/p_i}. \end{aligned}$$

This is discrete concavity in s ; input interchange gives the other coordinate. Applying (6) to the three-sector degeneration in Lemma 2.1 gives (9). $\square$

3 Hermite degeneration and ordinary scaling limits

The following degeneration makes each character profile superadditive. It will give degree-one limits directly, with the original dot-product exponents p_i .

Lemma 3.1 (Hermite direct sums). *For any nonempty finite list of positive integer pairs (r_h, s_h) , with $R = \sum_h r_h$ and $S = \sum_h s_h$,*

$$C_{ij}(R, S) \supseteq \bigoplus_h C_{ij}(r_h, s_h). \quad (10)$$

The summands are disjoint on all three legs.

Proof. It suffices to use the input legs X, Y . Choose distinct complex numbers $\alpha_1, \dots, \alpha_m$. Hermite interpolation gives a basis $u_{h,i}(t)$ for polynomials of degree less than R , specified by

$$u_{h,i}(t) \equiv \begin{cases} (t - \alpha_h)^i, & \ell = h, \\ 0, & \ell \neq h, \end{cases} \pmod{(t - \alpha_\ell)^{r_\ell}}, \quad 0 \leq i < r_h.$$

Choose the analogous basis $v_{h,j}(t)$ for the second input, using the orders s_h . For the output polynomial $W(t)$, retain the coefficients $[z^k]W(\alpha_h + z)$ for $0 \leq k \leq r_h + s_h - 2$; here $[z^k]$ means the coefficient of z^k . These $R + S - m$ values can be specified independently: Hermite interpolation realizes any such list with a polynomial of degree less than $R + S - m$, which fits the output degree bound $R + S - 1$.

In these fixed coordinates, scale each tensor variable by ε raised to its weight, where

$$\begin{aligned} w(x_{h,i}) &= r_h - 1 - i, & w(y_{h,j}) &= s_h - 1 - j, \\ w(z_{h,k}) &= k - r_h - s_h + 2. \end{aligned} \quad (11)$$

For two inputs from node h , the coefficient at $k = i + j$ equals one and has total weight zero. Any other nonzero coefficient at that node has $k > i + j$ and positive weight. If the first input comes from another node and the second is $v_{h,j}$, a nonzero coefficient requires $k \geq r_h + j$. Its weight is at least one, since the first-input weight is nonnegative. The reversed case is identical. If both inputs come from other nodes, their product vanishes to order at least $r_h + s_h$, beyond the retained output range.

Thus every nonzero term has nonnegative weight, and the limit as $\varepsilon \rightarrow 0$ is exactly the direct sum in (10). $\square$

Character additivity and degeneration monotonicity now give

$$f_{ij}(r + u, s + v) \geq f_{ij}(r, s) + f_{ij}(u, v). \quad (12)$$

Taking m copies of $C_{ij}(1, 1)$ also gives $f_{ij}(m, m) \geq m$; by monotonicity, $f_{ij}(r, s) \geq \min\{r, s\}$.

Lemma 3.2 (Ordinary homogeneous limits). *For every $r, s > 0$, the limit*

$$g_{ij}(r, s) = \lim_{n \rightarrow \infty} \frac{f_{ij}(\lfloor nr \rfloor, \lfloor ns \rfloor)}{n} \quad (13)$$

exists, uniformly for $r, s \in [u, v]$ for every fixed $0 < u < v$. The profiles g_{ij} are nondecreasing and concave, and are homogeneous of degree one, meaning

$$g_{ij}(tr, ts) = t g_{ij}(r, s) \quad (t > 0).$$

They also satisfy

$$\min\{r, s\} \leq g_{ij}(r, s) \leq r + s, \quad g_{ij}(r, s) = g_{ji}(s, r). \quad (14)$$

For $p_i > 0$, g_{ij}^{1/p_i} is concave in s and satisfies

$$g_{ij}(r, 3s + r)^{1/p_i} \geq 2g_{ij}(r, s)^{1/p_i} + g_{ik}(r, s)^{1/p_i}. \quad (15)$$

For $p_j > 0$, g_{ij}^{1/p_j} is concave in r .

Proof. Fix positive integers r, s . By (12), the sequence $a_n = f_{ij}(nr, ns)$ is superadditive and bounded above by $n(r + s)$. For each fixed k , writing $n = qk + t$, $0 \leq t < k$, gives $a_n \geq qa_k$. Therefore

$$\liminf_{n \rightarrow \infty} \frac{a_n}{n} \geq \frac{a_k}{k}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{n} = \sup_{k \geq 1} \frac{a_k}{k}.$$

This is the superadditive-sequence argument, also called Fekete's lemma.

For positive rational $r = a/q$, $s = b/q$, monotonicity squeezes the floored pair in (13) between $\lfloor n/q \rfloor(a, b)$ and $(\lfloor n/q \rfloor + 1)(a, b)$. Hence the limit is $g_{ij}(a, b)/q$. Superadditivity passes to rational pairs, and rational homogeneity makes the limit concave there.

To handle real lengths, first bound how much the limit can change between nearby rational points. Fix rational $0 < u < v$ and write $g = g_{ij}$. For rational $r, r', s \in [u, v]$ with $r < r'$, concavity gives

$$0 \leq \frac{g(r', s) - g(r, s)}{r' - r} \leq \frac{g(r, s) - g(u/2, s)}{r - u/2} \leq \frac{4v}{u},$$

using $0 \leq g(r, s) \leq r + s \leq 2v$. The same bound holds in the other coordinate. Thus changing the two coordinates by a total of η changes g by at most $4v\eta/u$. Approximating real lengths by rationals therefore defines a unique continuous extension of g .

Now cover any fixed rectangle with a fine rational grid. The slope bound makes the limit values at neighboring corners arbitrarily close. Both the normalized discrete profiles and g lie between their values at the lower and upper grid corners. Convergence at the finitely many corners therefore gives uniform convergence throughout the rectangle. Homogeneity, monotonicity, input interchange, and concavity, as well as (14), pass from rational to real lengths by continuity.

For powered concavity, first take rational lengths and a rational interpolation coefficient. Clear denominators and apply the discrete concavity from Lemma 2.2 along multiples of that common denominator. Divide by n^{1/p_i} and pass to the limit; continuity gives all real lengths and coefficients. The other coordinate is identical. Finally, local uniform convergence absorbs the bounded rounding errors and the -1 in (9), giving (15). $\square$

4 Common slopes and the angular constraint

Index the three tensor legs by 1, 2, 3. We use the limits above to constrain their dot-product exponents without symmetrizing them.

Lemma 4.1 (Common slope and intercepts). *Fix i with $p_i > 0$, and let j, k be the other legs. The functions $P_{ij}(s) = g_{ij}(1, s)^{1/p_i}$ and $P_{ik}(s) = g_{ik}(1, s)^{1/p_i}$ have a common asymptotic slope $L_i > 0$. Their intercepts*

$$C_{ij} = \lim_{s \rightarrow \infty} (P_{ij}(s) - L_i s), \quad C_{ik} = \lim_{s \rightarrow \infty} (P_{ik}(s) - L_i s)$$

are finite and nonnegative, with $C_{ij} + C_{ik} \leq L_i$.

Proof. Write $P = P_{ij}$ and $Q = P_{ik}$. Positive nondecreasing concavity gives finite nonnegative limits $L_P = \lim P(s)/s$ and $L_Q = \lim Q(s)/s$. Dividing the coupled inequalities

$$P(3s + 1) \geq 2P(s) + Q(s), \quad Q(3s + 1) \geq 2Q(s) + P(s)$$

by s and passing to infinity gives $L_P \geq L_Q$ and $L_Q \geq L_P$. Call their common value L_i .

Their sum $R = P + Q$ satisfies $R(3s + 1) \geq 3R(s)$. Iterating at $s_m = 3^m(s + \frac{1}{2}) - \frac{1}{2}$ gives

$$R(s) \leq \lim_{m \rightarrow \infty} 3^{-m} R(s_m) = L_i(2s + 1). \quad (16)$$

Since $R(s) > 0$, we have $L_i > 0$. Concavity makes the derivatives of P, Q decrease to L_i , so $P(s) - L_i s$ and $Q(s) - L_i s$ are nondecreasing and have nonnegative limits at zero. Their sum is bounded above by L_i by (16). Their limits at infinity therefore exist and have the stated bounds. $\square$

Proposition 4.2 (Angular character constraint). *Every tensor character with dot-product exponents p_1, p_2, p_3 satisfies*

$$\sum_{i=1}^3 \arccos \sqrt{p_i} \geq \frac{\pi}{2}. \quad (17)$$

Proof. If some $p_i = 0$, its summand is already $\pi/2$. Assume all three exponents are positive. Set $c_{ij} = C_{ij}/L_i$. Lemma 4.1 gives the six intercept ratios and their three bounds

$$c_{ij} \geq 0, \quad c_{ij} + c_{ik} \leq 1. \quad (18)$$

Also $P'_{ij} \geq L_i$ almost everywhere and $P_{ij}(s) \leq L_i(s + c_{ij})$, so

$$\frac{d}{ds} \log g_{ij}(1, s) = p_i \frac{P'_{ij}(s)}{P_{ij}(s)} \geq \frac{p_i}{s + c_{ij}}.$$

By homogeneity and input interchange, at differentiability points

$$\partial_s \log g_{ij}(r, s) \geq \frac{p_i}{s + c_{ij}r}, \quad \partial_r \log g_{ij}(r, s) \geq \frac{p_j}{r + c_{ji}s}.$$

The profiles are locally Lipschitz and hence differentiable almost everywhere. At such a point, differentiating $g_{ij}(tr, ts) = t g_{ij}(r, s)$ at $t = 1$ and dividing by $g_{ij}(r, s)$ gives

$$r \partial_r \log g_{ij}(r, s) + s \partial_s \log g_{ij}(r, s) = 1.$$

Combining the two derivative bounds and using continuity to include all positive r, s gives

$$1 \geq \frac{p_i s}{s + c_{ij}r} + \frac{p_j r}{r + c_{ji}s} \quad (r, s > 0). \quad (19)$$

With $t = r/s$, this is equivalent to

$$(1 - p_j)c_{ij}t^2 + (1 - p_i - p_j + c_{ij}c_{ji})t + (1 - p_i)c_{ji} \geq 0 \quad (t > 0).$$

A quadratic $At^2 + Bt + C$ with $A, C \geq 0$ is nonnegative on $t > 0$ exactly when $B \geq -2\sqrt{AC}$, including zero coefficients. Thus

$$\sqrt{c_{ij}c_{ji}} \geq K_{ij} := \left(\sqrt{p_i p_j} - \sqrt{(1 - p_i)(1 - p_j)} \right)_+, \quad (20)$$

where $(z)_+ = \max\{z, 0\}$.

Let K be the symmetric matrix with these off-diagonal entries and zero diagonal. For every vector $x \geq 0$, AM-GM and (18) give

$$x^\top K x \leq \sum_{i < j} 2\sqrt{c_{ij}c_{ji}}x_i x_j \leq \sum_{i < j} (c_{ij}x_i^2 + c_{ji}x_j^2) \leq \sum_i x_i^2. \quad (21)$$

Put $\theta_i = \arccos \sqrt{p_i}$. If $\sum_i \theta_i < \pi/2$, increase all three angles by the same positive amount to obtain $\phi_i > 0$ with sum $\pi/2$. The matrix K' with zero diagonal and $K'_{ij} = \cos(\phi_i + \phi_j) = \sin \phi_k$ is strictly smaller than K off the diagonal. But the positive vector $x_i = \cos \phi_i$ satisfies

$$(K'x)_i = \sin \phi_k \cos \phi_j + \sin \phi_j \cos \phi_k = \cos \phi_i = x_i,$$

contradicting (21). $\square$

5 Matrix-multiplication exponents

We first optimize the angular constraint for three arbitrary dimensions; the rectangular curve then follows by substitution.

Corollary 5.1. *For every fixed $a \geq b \geq c \geq 0$,*

$$\omega(a, b, c) \leq \begin{cases} a + b, & c(a + b) \leq ab, \\ \frac{a + b + c}{2} + \frac{1}{4} \left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a} \right), & c(a + b) > ab. \end{cases} \quad (22)$$

When $c(a + b) \leq ab$, equality holds by the size lower bound.

Proof. Reorder the dot-product exponents so that the character objective is $ap_1 + bp_2 + cp_3$, and write $p_i = \cos^2 \theta_i$ with $\theta_i \in [0, \pi/2]$. By Proposition 4.2, their sum is at least $\pi/2$. Since the objective decreases with each angle, its maximum over this region occurs on $\sum_i \theta_i = \pi/2$. If a weight is zero, the maximum is $a + b$. Assume $a, b, c > 0$.

On any boundary edge, the other two angles sum to $\pi/2$, so their contribution is a convex combination of the corresponding two weights. The largest boundary value is therefore $a + b$. At an interior stationary point, the multiplier equations are

$$a \sin(2\theta_1) = b \sin(2\theta_2) = c \sin(2\theta_3).$$

The angles $2\theta_i$ form a triangle with sides proportional to $1/a, 1/b, 1/c$. Such a triangle exists uniquely exactly when $1/c < 1/a + 1/b$, or $c(a + b) > ab$. The cosine rule gives

$$\cos(2\theta_1) = \frac{c}{2b} + \frac{b}{2c} - \frac{bc}{2a^2},$$

and cyclic variants. Using $p_i = (1 + \cos(2\theta_i))/2$ yields the second branch of (22). It exceeds $a + b$ by $(c(a + b) - ab)^2 / (4abc) > 0$. Thus it is the global maximum when the triangle exists; otherwise the maximum is the boundary value $a + b$. At $c(a + b) = ab$, the branches agree.

For rational shapes, choose an integer $q > 1$ with q^a, q^b, q^c integral. By (3) and (4), the bound just proved is an upper bound on $\log_q \text{rank}(\langle q^a, q^b, q^c \rangle)$. Choose a tensor power whose rank is within arbitrary exponent slack, and recurse with its fixed bilinear decomposition. The rank and flattening dimensions cover the additions, and padding to arbitrary sizes preserves the exponent. This gives the claimed running time. For real shapes, approximate each exponent from above by rationals and pad. The optimized bound is continuous in the weights, as the maximum of these linear objectives over a fixed compact set, completing the proof. $\square$

Proof of Theorem 1.1. Apply Corollary 5.1 to the sorted dimensions of $(1, k, 1)$. For $0 \leq k \leq 1$, the first branch applies exactly when $2k \leq 1$. In the remaining range, both possible orderings give

$$\frac{2 + k}{2} + \frac{1}{4} \left(2k + \frac{1}{k} \right) = 1 + k + \frac{1}{4k}.$$

The output-size lower bound $\omega(k) \geq 2$ proves equality for $0 \leq k \leq \frac{1}{2}$ and hence $\alpha \geq \frac{1}{2}$. $\square$

Some values of the bound in Theorem 1.1 are:

k	0.5	0.6	0.8	1	1.5	2
$\omega(k) \leq$	2	2.0166...	2.1125	2.25	2.6666...	3.125

The exponent convention gives an $O(n^2)$ -time algorithm at $k = \frac{1}{2}$.

6 Applications

Theorem 1.1 and Corollary 5.1 give the following bounds through standard rectangular-multiplication balances. All bounds use arithmetic operations, with graph operations and symbol comparisons also charged.

1. **One lowest common ancestor per pair in a DAG.** Grandoni et al. [5, Theorem 4.2 of the published version] obtain exponent $1+2x$ from the balance $\omega(x) = 3x$. Theorem 1.1 permits $x = (1+\sqrt{3})/4$, giving $O(n^{(3+\sqrt{3})/2})$ time for finding one lowest common ancestor for every pair in an n -vertex DAG. The exponent is $(3 + \sqrt{3})/2 \approx 2.36603$.
2. **Exact all-pairs Hamming distance.** For n vectors of length n over an unrestricted alphabet, the reduction to dominance products [7] and Yuster’s tradeoff [10, Proposition 2.1] give the balance $\rho = \omega(4 - \rho)$. Theorem 1.1 gives $\rho = (13 - \sqrt{7})/4 \approx 2.58856$ and running time $O(n^\rho)$ for computing every exact Hamming distance.
3. **The standard unweighted APSP balance.** Zwick’s deterministic algorithm for directed unweighted APSP [12, Section 6] has exponent $2 + \mu$, where $\omega(\mu) \leq 1 + 2\mu$. Here $\omega(\frac{1}{2}) = 2 = 1 + 2(\frac{1}{2})$, so it gives $O(n^{2.5})$ running time. Combining the full curve with the deterministic (min, +)-product algorithm and APSP reduction of Alman and Vassilevska Williams [1, Theorems 22 and 33–34] further gives $O(n^{2.4999})$ running time.
4. **Minimum and maximum witnesses.** For two Boolean $n \times n$ matrices, a minimum (maximum) witness for entry (i, j) is the least (greatest) index k with $A_{ik} = B_{kj} = 1$. The standard blocking algorithm [4, Section 2.1] takes $O(n^{1-r+\omega(r)} + n^{2+r})$ time. Taking $r = \frac{1}{2}$ gives $O(n^{2.5})$ time for all witnesses.
5. **Sparse matrix multiplication.** For two $n \times n$ matrices with at most m nonzero entries each, the sparse–dense decomposition of Yuster and Zwick [11, Section 3] gives $O(m^2/n^r + n^{\omega(r)})$ time for $0 \leq r \leq 1$. Thus $m \leq n^{1.25}$ permits $O(n^2)$ time.
6. **Dynamic matrix inverse and reachability.** Substituting Corollary 5.1 into van den Brand–Nanongkai–Saranurak’s tradeoff [9, Theorem 4.2], with $u = 0.50323551$ and $v = 0.80667763$ in $\max\{u + v, \omega(1, u, v) - u, \omega(v) - v\}$, gives $O(n^{1.30992})$ worst-case time per entry update and query for maintaining the inverse of a nonsingular $n \times n$ matrix, after $O(n^{2.25})$ preprocessing. The associated Monte Carlo reduction gives the same bounds for directed reachability under edge updates, with $O(1)$ query time for a fixed pair [9, Corollary C.12 and Theorem C.16].

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A The determinant construction

We recap the determinant construction used in [8, Lemma 4.1] as an explicit change of basis for pairs of univariate polynomials. Its classical form is the degree-one Clebsch–Gordan formula [6, Theorem 2.6.3, pp. 56–57 of the author version].

Fix $r \geq 1$ and $s \geq 2$. The tensor $C_{XY}(r, s) \otimes B_X(2)$ computes

$$(f, (a, b)) \longmapsto (fa, fb), \quad \deg f < r, \quad \deg a, \deg b < s.$$

Thus tensoring with $B_X(2)$ means multiplying two polynomials by the same f . We will choose fixed bases in which this computation splits, in the limit, into multiplication with input lengths $s + 1$ and $s - 1$.

For $e \geq 1$, consider pairs (a, b) with $\deg a, \deg b \leq e$. The polynomial $a + tb$ has degree at most $e + 1$, and

$$a + tb = 0 \iff (a, b) = (-th, h) \text{ for some } \deg h \leq e - 1. \quad (\text{A.1})$$

Use the following basis, arranged in two blocks:

$$\underbrace{(t^j, 0) \ (0 \leq j \leq e), \ (0, t^e)}_{e+2 \text{ vectors}}, \quad \underbrace{(-t^{j+1}, t^j) \ (0 \leq j < e)}_{e \text{ vectors}}.$$

The first block maps under $(a, b) \mapsto a + tb$ to $1, t, \dots, t^{e+1}$. Every pair mapping to zero is a unique linear combination of the second block, by (A.1). These vectors therefore form a basis. At $e = s - 1$, the two blocks have sizes $s + 1$ and $s - 1$.

Choose these bases for both the second-input pairs ($e = s - 1$) and the output pairs ($e = r + s - 2$). Multiplication by every f satisfies

$$fa + t(fb) = f(a + tb), \quad f(-th, h) = (-t(fh), fh).$$

The first identity says that the first block undergoes ordinary multiplication by f ; the second says the same for the second block and shows that it contributes nothing to the first. Consequently, the matrix of multiplication by f has the form

$$\begin{pmatrix} M_f^+ & 0 \\ R_f & M_f^- \end{pmatrix}, \quad \begin{matrix} M_f^+ : \mathbb{C}^{s+1} \longrightarrow \mathbb{C}^{r+s}, \\ M_f^- : \mathbb{C}^{s-1} \longrightarrow \mathbb{C}^{r+s-2}, \end{matrix} \quad (\text{A.2})$$

where M_f^+, M_f^- are the usual coefficient matrices for multiplying by f . The bases are fixed independently of f , so this block structure holds for the whole tensor.

Scale the variables in the second block on the second-input leg by ε^{-1} , and those in the second block on the output leg by ε . On every multiplication matrix this gives

$$\begin{pmatrix} I & 0 \\ 0 & \varepsilon I \end{pmatrix} \begin{pmatrix} M_f^+ & 0 \\ R_f & M_f^- \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & \varepsilon^{-1} I \end{pmatrix} = \begin{pmatrix} M_f^+ & 0 \\ \varepsilon R_f & M_f^- \end{pmatrix} \xrightarrow{\varepsilon \rightarrow 0} \begin{pmatrix} M_f^+ & 0 \\ 0 & M_f^- \end{pmatrix}.$$

The two surviving tensor blocks are $C_{XY}(r, s + 1)$ and $C_{XY}(r, s - 1)$. They share the first leg and have disjoint second and third legs. Permuting the legs gives the other orientations.

Finally, for $p_i > 0$, multiplicativity of λ , the identity $\lambda(B_i(2)) = 2^{p_i}$, degeneration monotonicity, and (6) give

$$\begin{aligned} 2\lambda(C_{ij}(r, s))^{1/p_i} &= \lambda(C_{ij}(r, s) \otimes B_i(2))^{1/p_i} \\ &\geq \lambda(C_{ij}(r, s + 1))^{1/p_i} + \lambda(C_{ij}(r, s - 1))^{1/p_i}. \end{aligned} \quad (\text{A.3})$$